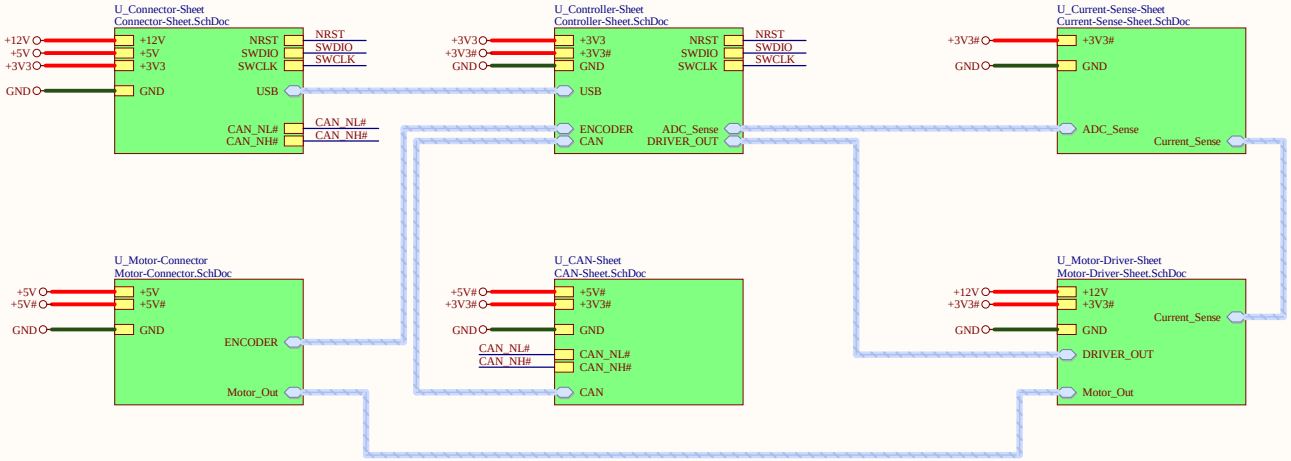
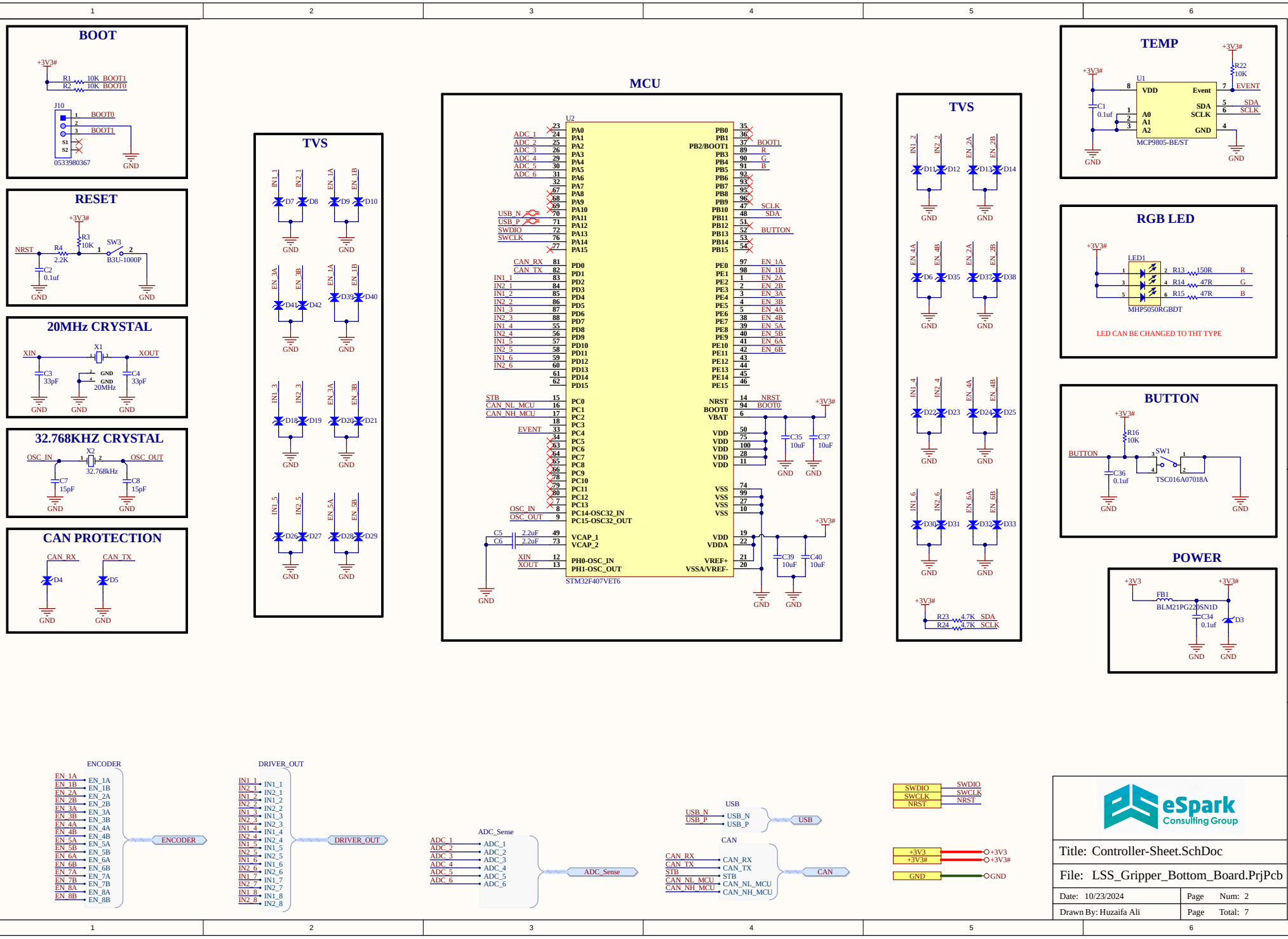
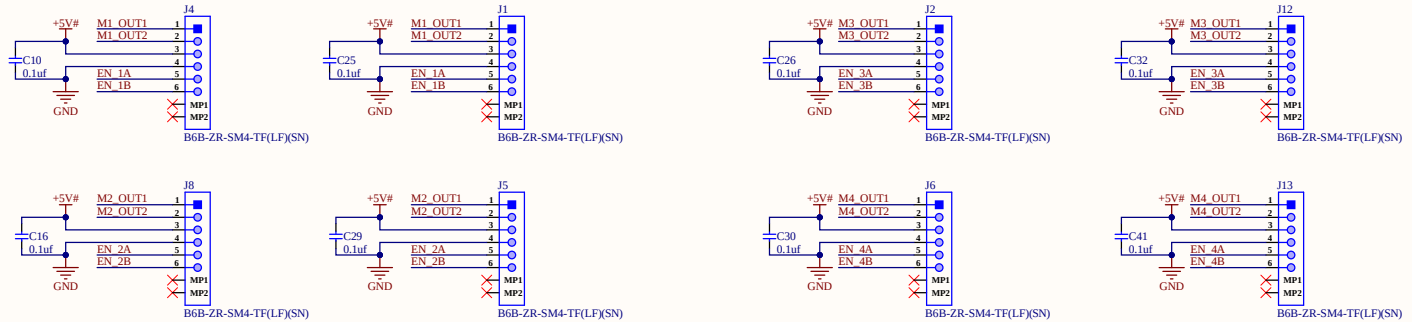


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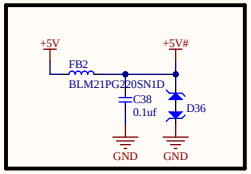
Title: Top-Sheet.SchDoc	
File: LSS_Gripper_Bottom_Board.PrjPcb	
Date: 6/7/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 7


BOOT
+3V3#
R1 10K BOOT1
R2 10K BOOT0
J10
BOOT0
BOOT1
S1
S2
0533980367
GND
RESET
+3V3#
NRST
R4 2.2K
R3 10K
SW3
B3U-1000P
C2 0.1uf
GND
20MHz CRYSTAL
X1
XIN
XOUT
C3 33pF
C4 33pF
GND
20MHz
32.768KHZ CRYSTAL
X2
OSC_IN
OSC_OUT
32.768kHz
C7 15pF
C8 15pF
CAN PROTECTION
CAN_RX
CAN_TX
D4
D5
TVS
IN1_1
IN2_1
EN_1A
EN_1B
D7
D8
D9
D10
EN_3A
EN_3B
D41
D42
D39
D40
IN1_3
IN2_3
EN_2A
EN_2B
D18
D19
D20
D21
IN1_5
IN2_5
EN_5A
EN_5B
D26
D27
D28
D29
MCU
U2
ADC_1
ADC_2
ADC_3
ADC_4
ADC_5
ADC_6
USB_N
USB_P
SWDIO
SWCLK
CAN_RX
CAN_TX
STB
CAN_NL_MCU
CAN_NH_MCU
EVENT
IN1_1
IN2_1
IN1_2
IN2_2
IN1_3
IN2_3
IN1_4
IN2_4
IN1_5
IN2_5
IN1_6
IN2_6
OSC_IN
OSC_OUT
C5 2.2uF
C6 2.2uF
XIN
XOUT
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
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PD10
PD11
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PD14
PD15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
VCAP_1
VCAP_2
PH0-OSC_IN
PH1-OSC_OUT
STM32F407VET6
PB0
PB1
PB2/BOOT1
PB3
PB4
PB5
PB6
PB7
PB8
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
NRST
BOOT0
VBAT
VDD
VSS
VDDA
VREF+
VSSA/VREF-
BOOT1
R
G
B
SCLK
SDA
BUTTON
EN_1A
EN_1B
EN_2A
EN_2B
EN_3A
EN_3B
EN_4A
EN_4B
EN_5A
EN_5B
EN_6A
EN_6B
C35 10uF
C37 10uF
C39 10uF
C40 10uF
TVS
IN1_2
IN2_2
EN_2A
EN_2B
D11
D12
D13
D14
EN_4A
EN_4B
D6
D35
D37
D38
IN1_4
IN2_4
EN_4A
EN_4B
D22
D23
D24
D25
IN1_6
IN2_6
EN_6A
EN_6B
D30
D31
D32
D33
R23 4.7K SDA
R24 4.7K SCLK
TEMP
U1
VDD
Event
A0
A1
A2
GND
SDA
SCLK
EVENT
R22 10K
C1 0.1uf
MCP9805-BE/ST
RGB LED
LED1
R13 150R R
R14 47R G
R15 47R B
MHP5050RGBDT
LED CAN BE CHANGED TO THT TYPE
BUTTON
R16 10K
SW1
C36 0.1uf
TSC016A07018A
POWER
+3V3
FB1
BLM21PG220SN1D
C34 0.1uf
D3
ENCODER
EN_1A
EN_1B
EN_2A
EN_2B
EN_3A
EN_3B
EN_4A
EN_4B
EN_5A
EN_5B
EN_6A
EN_6B
EN_7A
EN_7B
EN_8A
EN_8B
DRIVER_OUT
IN1_1
IN2_1
IN1_2
IN2_2
IN1_3
IN2_3
IN1_4
IN2_4
IN1_5
IN2_5
IN1_6
IN2_6
IN1_7
IN2_7
IN1_8
IN2_8
ADC_Sense
USB
CAN
SWDIO
SWCLK
NRST
+3V3
GND
eSpark
Consulting Group
Title: Controller-Sheet.SchDoc
File: LSS_Gripper_Bottom_Board.PrjPcb
Date: 10/23/2024
Page Num: 2
Drawn By: Huzaifa Ali
Page Total: 7

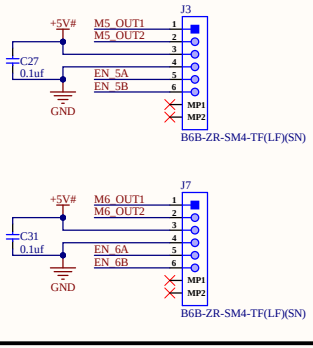
MOTOR CONNECTORS

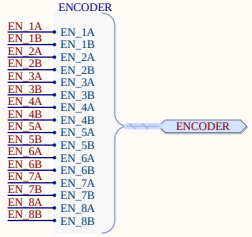


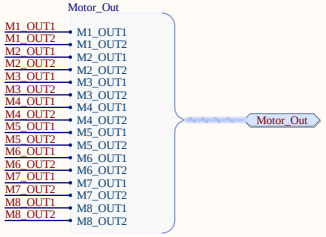
POWER

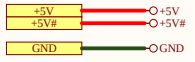


ADDITIONAL CONNECTORS









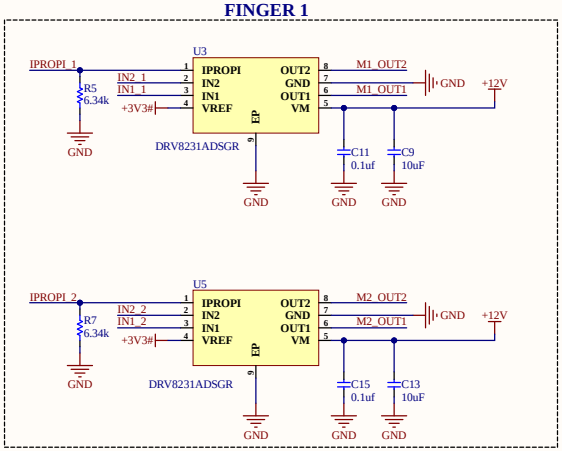
1. Black = - Motor
2. Red = + Motor
3. Brown = Hall Sensor Vcc
4. Green = Hall Sensor Gnd
5. Blue = Sensor A
6. Yellow = Sensor B

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Title: Motor-Connector.SchDoc	
File: LSS_Gripper_Bottom_Board.PrjPcb	
Date: 11/5/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 7

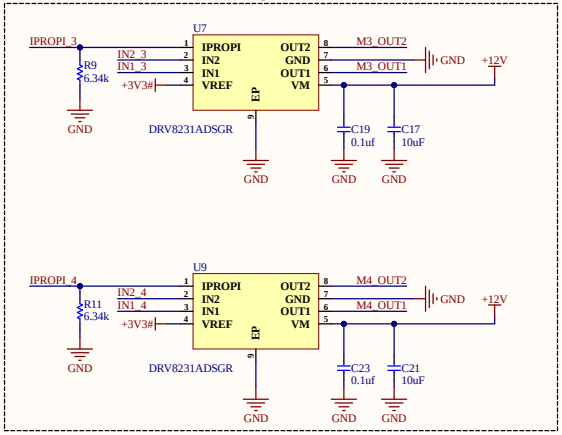
STEPPER DRIVER

FINGER 1

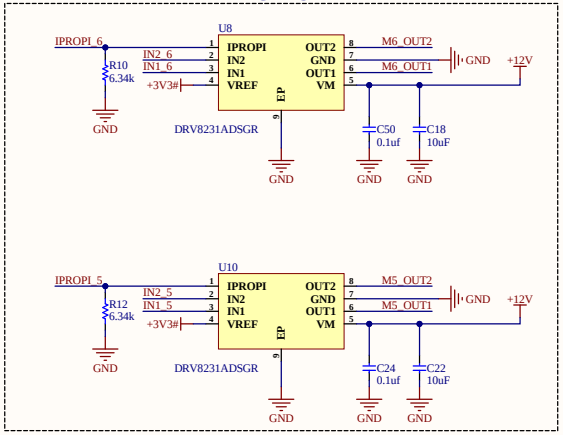


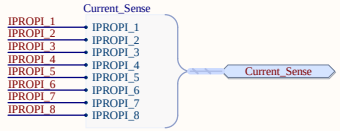
Resistor based on 347mA STALL Current = 6.34K

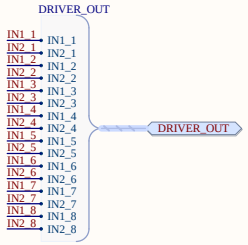
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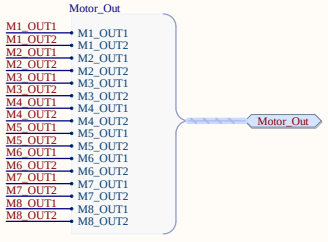


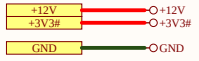
FINGER 3





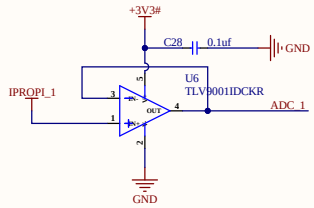


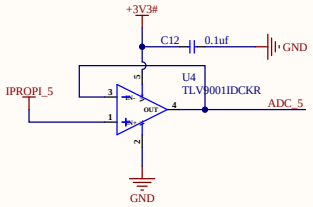


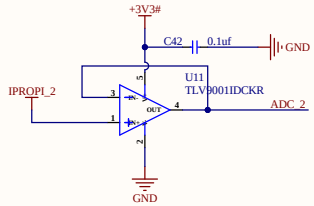


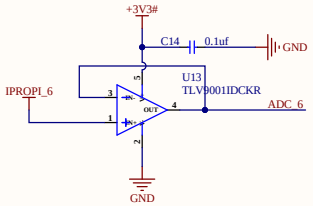
eSpark Consulting Group

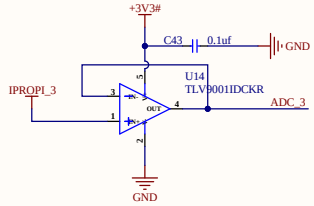
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File: LSS_Gripper_Bottom_Board.PrjPcb	
Date: 10/14/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 7

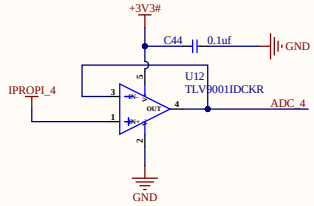


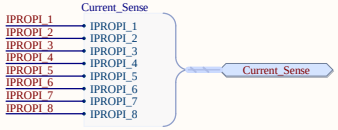


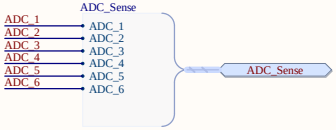








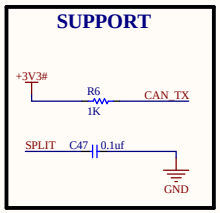


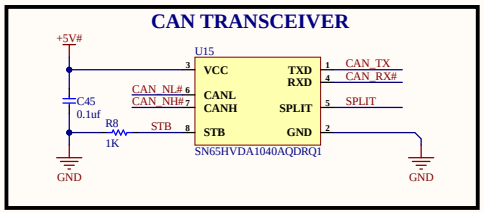


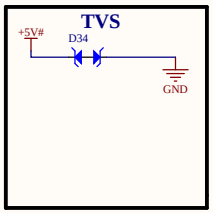


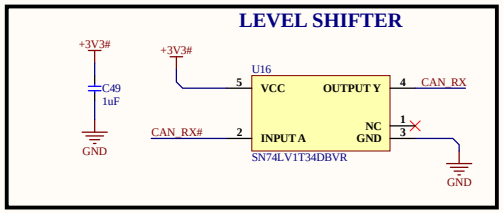
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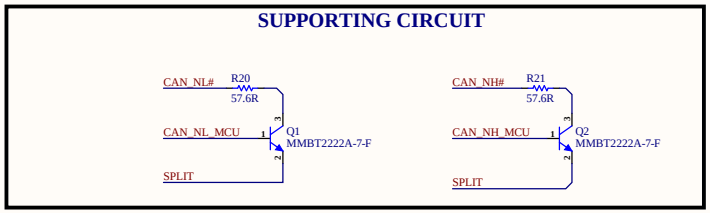
Title: Current-Sense-Sheet.SchDoc	
File: LSS_Gripper_Bottom_Board.PrjPcb	
Date: 6/7/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 7

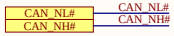












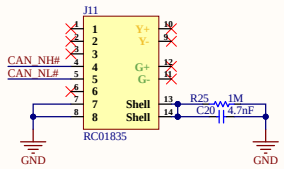




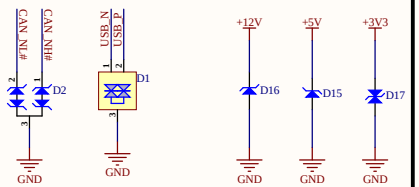
eSpark Consulting Group

Title: CAN-Sheet.SchDoc	
File: LSS_Gripper_Bottom_Board.PrjPcb	
Date: 10/14/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 7

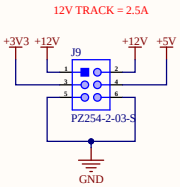
RJ45



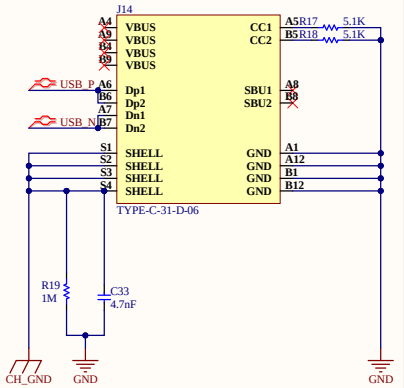
ESD PROTECTION



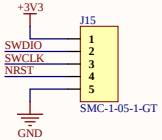
CONNECTOR

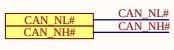


USB-C

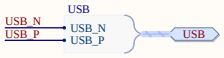


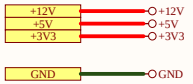
SWD











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Title: Connector-Sheet.SchDoc	
File: LSS_Gripper_Bottom_Board.PrjPcb	
Date: 10/23/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 7